



AI in remote sensing and satellite image processing-a review

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Abstract

The integration of artificial intelligence (AI) in remote sensing and satellite image processing has significantly transformed the field, offering advanced tools for data analysis, feature extraction, and environmental monitoring. With the growing availability of high-resolution satellite imagery, AI applications such as machine learning and deep learning have been applied to automate the process of interpreting complex spatial data. This paper intends to explore the current state of AI in remote sensing, focused on some applications such as land cover classification, object detection, climate change monitoring, and disaster management. Additionally, the challenges and future directions for AI-driven remote sensing are discussed, emphasizing the need for better generalizability, data fusion, and improved computational efficiency.

Keywords Artificial intelligence · Remote sensing · Satellite image processing · Machine learning · Deep learning · Environmental monitoring · Data fusion

Introduction

Remote sensing, using satellite or airborne sensors for the detailed investigation of the earth's surface, has become indispensable in tackling global and local problems, such as environmental conservation, precision agriculture, urban planning, and natural disaster response. Modern advancements in the technology of sensors have generated a massive amount of functional data, which has enlarged the variety and resolution of satellite imagery retrieval, thus enabling more accurate and extensive visibility monitoring of terrestrial, atmospheric, and marine environments (Chibuike and Nzeanorue 2024). However, with the booming out of data came additional burdens on managing, analyzing, and interpreting the newly drawn-out huge amounts in time and space (Zhipeng et al. 2023).

There is an AI approach to tackling the unique and very complex nature of remote sensing data. AI finds the solutions in machine learning (ML) and deep learning (DL) techniques, as they provide the power to automate the analyses of large geospatial datasets, the discovery of patterns, and the extraction of valuable information. By using sophisticated algorithms, AI enables understanding of subtle spatial, temporal, and spectral variations in satellite imagery that normally would go undetected by conventional methods (Rao et al. 2024). For example, through models built upon AI principles, urban expansion patterns can be detected, changes can be seen in land cover, crop health can be monitored, and devastation impacts can be assessed (Zhujun and Maimai 2023).

Machine learning techniques such as support vector machines (SVMs), random forests, and k-means clustering have been used in remote sensing for some time for land cover classification and vegetation monitoring (Firas et al. 2024; Asif et al. 2024). However, this new evolution has widened the capability of AI applications in remote sensing, as deep neural networks (DNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs) allow the models to learn complex data hierarchies as well as directly drive from raw imagery. Thus, deep learning models are used for better effectiveness in the analysis of high-resolution imagery and multidimensional data (e.g., multispectral or hyperspectral images) and for accuracy

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improvement in object detection, segmentation, and feature extraction (Tachasit et al. 2024).

This review paper highlights the wide-ranging applications of AI in remote sensing and satellite image processing, including the most recent developments and ongoing challenges, as well as the future potential of AI-centered remote sensing. We shall discuss different use cases, including environmental monitoring, where AI shapes outcomes on changes in deforestation, biodiversity losses, and impacts of climate change; agricultural processes that take advantage of an AI's potential to minimize crop stress, optimal irrigation, and yield prediction; and many other examples to come. The challenges with AI in remote sensing and satellite image processing will also be discussed and will be used to develop guidelines for future works.

Applications of AI in remote sensing

Land cover and land use classification

Land cover and land use classification are critical components of remote sensing, as they provide essential information for understanding environmental changes, urban development, agricultural activities, and ecological health (Firas et al. 2024). The application of AI has resulted in impressive advantages, enhancing the users' ability to effectively and efficiently classify tasks using deep learning and machine learning models.

Traditional methods for classifying land cover face severe obstacles from the complexity induced by the diversity of landscapes and the data volume available from satellites: maximum likelihood classifiers and support vector machines (SVMs) (Gadiparthi et al. 2024). Recently, deep learning, particularly through convolutional neural networks (CNNs), has emerged as a key method for addressing the problems arising from the preceding classifiers. With their high capacity for automatically learning spatial patterns from high-resolution images and extracting key features, CNNs are particularly effective when it comes to differentiating accurately between the various types of land covers.

Recent research further demonstrates that CNN-based models are now being applied beyond generic object recognition, with direct implications for environmental earth sciences. For example, CNN architectures have been successfully used to map deforestation patterns in tropical forests (Pozzobon de Bem et al. 2020), monitor vegetation stress through integration with spectral indices such as NDVI (Tufail et al. 2025). At the same time, several technical studies have highlighted their superior performance in urban and peri-urban environments. For instance, an interesting study demonstrated that the CNN-based method achieved

high accuracy in road extraction from high-resolution orthophoto images, with average precision, recall, F1 score, overall accuracy, and IOU percentages of 91.09%, 95.32%, 93.15%, 94.44%, and 87.21%, respectively (Abdollahi et al. 2021). In another work, the CNN model effectively classified aerial photographs into seven land cover classes, including roads, achieving an overall accuracy of 0.973 and a Kappa coefficient of 0.967, demonstrating its capability for accurate differentiation of various land cover types, including roads (Sameen et al. 2018). Hybrid architectures combining deep CNNs with random forests have also been applied in agricultural land classification, showing that integrating traditional ML with DL can enhance accuracy in heterogeneous regions. For instance, an interesting work proposed a CNN-Random Forest Hybrid model for phenology-based paddy rice mapping, achieving an overall accuracy of 0.950. This approach effectively integrates deep learning with traditional classification methods, enhancing agricultural land classification using multi-temporal satellite imagery (Sudiana et al. 2025). Together, these applications demonstrate that CNNs not only advance computational efficiency but also directly support biodiversity monitoring, urban expansion tracking, land degradation assessment, and sustainable land management.

Another approach that has gained traction is the use of transfer learning, which allows pre-trained deep learning models to be adapted for specific land cover classification tasks. This approach has proven particularly useful when labeled data are scarce, as it reduces the need for large, labeled training datasets (Matteo et al. 2024). A case in point is the study (Liu and Wang 2024) in which land cover classification in rural settings was undertaken with the help of a pre-trained ResNet and land cover classification through transfer learning, demonstrating a considerable enhancement even when using limited unlabeled data. This means that transfer learning might be a better alternative for developing countries or regions with very resource-intensive labeled data collection.

Attention mechanisms have been integrated into deep learning models to improve the precision in land cover classification. Attention mechanisms allow an emphasis on the most informative image regions, thereby facilitating better feature extraction (Zhipeng 2024). An instance is the work presented in Abhiroop et al. (2024), where the attention-based CNN model for land use classification was implemented, demonstrating that the attention module could greatly enhance the model in identifying subtle differences between similar land cover classes.

Random forests and gradient boosting machine models are currently in widespread use, particularly in projects with interpretability and simplicity demands (Giulia et al. 2023). These methodologies achieved successful classification of

agricultural and forested sites when combined with spectral indices like Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI) (Nayak and Panda 2024).

Recent years have seen an increase in the exploration of self-supervised and semi-supervised learning techniques to address the challenge of limited data. This form of learning gives the model knowledge of feature representations by using vast amounts of unlabeled satellite imagery, which is further fine-tuned for specific classification tasks (Jayanth et al. 2024). For instance, a study (Wenqing et al. 2024) used a self-supervised learning approach to land cover classification and showed improvements in the performance of deep learning models with land cover classification tasks on limited labeled data, thereby reducing the dependency on labor-intensive labeling efforts.

Overall, AI-driven land cover and land use classification have made some meaningful advances in accuracy, efficiency, and automation. The collision of deep learning with transfer learning, attention mechanisms, and self-supervised learning has transformed the remote sensing data processing, giving rise to more detailed and accurate assessments of environmental and urban dynamics.

Object detection and feature extraction

Object detection and feature extraction form two major applications of artificial intelligence in remote sensing—the accurate identification and classification of objects from satellite imagery. AI models have been significantly developed to detect various features like buildings, roads, vegetation, water bodies, and land-use patterns, which have all brought about benefits to urban planning, environmental monitoring, resource management, and more. Researchers (Valliappan et al. 2024), for example, had developed and applied convolutional neural networks (CNNs) to identify urban structures in high-resolution satellite images, whereby they achieved over 90% in classification tasks and further supported urban planning by detailed urban layouts. Likewise, coastal infrastructure monitoring was conducted by studies through the CNN-based object detection approach, which proved to be very sensitive in detecting changes. For instance, a study utilized a CNN to detect breaking points, crest orientation, and relative Still Water Level (SWL) in surfing waves, achieving a mean Average-Precision (mAP) of 0.794, demonstrating its effectiveness in monitoring dynamic coastal resources (Atkin et al. 2023). Another work implemented CNN-based semantic segmentation for monitoring coastal surface changes, achieving a global accuracy of 93.9% and an Intersection over Union (IoU) range of 86.6–92.7%, effectively detecting changes in classes like mangrove, vegetation, sand, and sea (Padilla-Arballo et al. 2022).

Traditional methods, such as edge detection and template matching, have limitations regarding handling the features of diversity, scale, orientation, and complexity in satellite imagery (Hewa et al. 2024). These limitations can be overcome by AI-based techniques, particularly through CNNs that automatically learn and derive spatial hierarchies. For instance, a study (Hou and Jing 2024) used R-CNN in detecting buildings and vehicles within the semiconfined spaces of very urbanized areas and reported remarkable improvements in precision against traditional methods. Other authors (Yash et al. 2024) used the YOLO (You Only Look Once) model to detect agricultural features like crop fields and irrigation systems. They concluded that YOLO's high speed and accuracy greatly favor it for large-scale agricultural monitoring. In addition, a recent study (Rajasekhar 2024) used YOLO to detect and identify various land cover types in satellite images, showing its potential for quick analyses in applications needing to cover large areas.

Feature extraction is a prerequisite and a significant component in converting raw satellite images into valuable information through quantifying shape, texture, spectral characteristics, and spatial relationships (Xizhen et al. 2024). For example, studies (Potić et al. 2023) have combined spectral indices such as the Normalized Difference Vegetation Index (NDVI) with machine learning models, including support vector machines (SVMs), to classify types of vegetation at high accuracy levels in forested and agricultural areas.

Recent innovations in AI bring attention mechanisms and multi-task learning frameworks that aim at improving object detection performance in remote sensing applications (Danqing and Yiquan 2023; Sadique et al. 2024). For example, researchers (Ashen and Hewarathna 2024) developed an attention-based CNN model for monitoring deforestation patterns, such that the attention mechanism focuses on regions that are integral to improving the accuracy of classification for expansive, heterogeneous forest landscapes.

Transfer learning has also gained popularity, as it enables models pre-trained on large datasets to be adapted to specific remote sensing tasks with minimal additional training (Yuan et al. 2023; Albughdadi et al. 2024). Researchers (Feng and Yuan 2024) have employed transfer learning using fine-tuning in region-specific datasets, demonstrating the improvement in accuracy in distinguishing among urban, suburban, and rural landscapes. Another study found that a ResNet50 model pre-trained on the RESISC-45 dataset achieved the highest accuracy of 97.10% when classifying the Eurosat dataset, demonstrating the effectiveness of transfer learning in remote sensing applications (Fahed 2022).

To sum up, advancements in artificial intelligence for object detection and feature extraction are changing the scenario of remote sensing by analyzing satellite images more

accurate and effective. The monitored detail can be done on the environmental and human-made constructions, as the value of these techniques increases from sustainable development to disaster management applications.

Climate change monitoring

Climate change monitoring is a significant application of AI in remote sensing that provides a better understanding of environmental changes, greenhouse gas emissions, land-use changes, and natural resource dynamics (Chuanming 2024). These tools offer vast amounts of data collected using sensors, and this data can be used for the assessment and mitigation of the impacts of climate change. For example, deep learning techniques have been used to observe the snow cover changes in mountain regions by adding it as an important parameter to trace water resources demarcated by global warming (Pinder 2024).

For monitoring climate, the conventional methods utilize manual data analysis and statistical models, which usually confine the spatial–temporal complexities of data. With AI models, especially deep learning, large volumes of complex data can be assessed, analyzed, and processed more accurately. For example, one work estimated the seasonal CO₂ emission in both urban and rural settings by applying satellite data through recurrent neural network-based analysis, thus providing information on anthropogenic impacts on greenhouse gas levels (Zhonghua et al. 2024). Another work espoused real-time monitoring on methane emissions through an AI model that integrated satellite observations with analytic predictions to track methane sources and trends in the agricultural and industrial sectors (Anna, et al. 2024).

The health of forests and land-use changes significantly impact climate monitoring, as they are the primary sources of greenhouse gas emissions and biodiversity loss (Nicole et al. 2024). One study established deep learning-based object detection algorithms to monitor illegal logging and forest degradation in the Amazon rainforest with over 90% accuracy (Gupta et al. 2024). Another research investigated random forest and spectral indices like NDVI for monitoring tropical deforestation to generate the necessary actionable data for conservation and reforestation (Ana et al. 2023).

AI models are also instrumental in tracking oceanic and atmospheric conditions to better understand sea level rise, ocean temperature variations, and extreme weather patterns. Deep neural networks were applied by researchers to analyze satellite data for sea level rise and coastal erosion, thus generating maps of flood-prone areas with good accuracy (Khadar et al. 2024). Another use of machine learning was to monitor coral bleaching and marine ecosystem health in response to rising ocean temperature, thus aiding

early warning systems for biodiversity protection (Narciso et al. 2024). In yet another study, researchers applied the long short-term memory (LSTM) model to analyze ocean temperature data for predicting marine heatwave events (Qi et al. 2024).

In addition, climate-driven events such as wildfires, droughts, and floods are monitored and forecasted through AI-based predictive models, which are fundamental in disaster preparedness. For example, machine learning was used to predict the risk of wildfires with the incorporation of weather and land-use datasets, such as air temperature, vapor pressure deficit, NDVI, and fuel moisture, attaining 72% accuracy with the most significant predictors for improved predictions (Heechan et al. 2024). Data of this type can be useful for managing water resources, as noted in another possible drought prediction model (Haitham and Afan 2024). In addition, floods were predicted through deep learning trained on rainfall and river flow data for enhanced disaster management in flood-prone regions (Jamunadevi et al. 2024). An interesting study utilized deep learning models, specifically GRU and LSTM, integrating upstream river flow, river water level, and tidal level data to enhance flood prediction accuracy, supporting effective disaster management in flood-prone regions like the Pattani River basin (Duangkhan 2025). Another work discussed an AI-based framework for multi-risk assessment in the Veneto region, integrating single hazard susceptibility maps and socio-economic vulnerability indicators to enhance understanding of extreme climate events, particularly focusing on heatwaves, droughts, and their impacts on tourism and agriculture (Ferrario et al. 2025).

In short, AI has transformed climate change monitoring through large-scale, automated analysis of environmental changes. With machine learning, deep learning, and predictive modeling combined, researchers can now monitor climate indicators across different sectors and topics. These developments translate into better-informed decisions supportive of climate resilience, conservation, and environmental policy, contributing toward global sustainability.

Disaster management

AI has significantly enhanced disaster management by systematically analyzing vast volumes of real-time remote sensing datasets, enabling speedy mitigation responses to chronic events such as earthquakes, floods, wildfires, hurricanes, and landslides. AI models help to emphasize the level of destruction, predict appropriate locations for more disasters, and ensure the proper distribution of resources during crises. For instance, this study (Guangxing and Gwanghyun 2024) utilized convolutional neural networks (CNNs) for the analysis of objects (like buildings and infrastructural

objects) in hurricane-hit areas, identifying destroyed buildings and infrastructure, which enabled efficient relief efforts in the most affected areas. Likewise, in another study (Furkan et al. 2024), deep learning was employed with post-disaster satellite imagery to assess the damage following an earthquake, thereby increasing the classification of damage via automatic means.

Classical disaster management stands to be rudimentary, being persistently dependent on manual assessment and scant analytical capacities (Shailen 2023; Grimaz et al. 2023). Thus, called deep learning, the AI approach proves to be more prompt as well as accurate. Similarly, some of the latest research used machine learning to predict flood risks given topographic and meteorological data, which would assist in the precision prediction of regions to prioritize in responding to a flood. For instance, a study (Shrestha et al. 2025) utilized machine learning algorithms, including logistic regression, to analyze topographical and meteorological data for flood risk assessment in Charlotte, North Carolina, effectively identifying and prioritizing regions for emergency preparedness and infrastructure planning based on flood susceptibility. Another work utilized machine learning models, incorporating 13 explanatory variables from topographical, hydrological, and environmental data, to create flood susceptibility maps for Thessaly, Greece, enhancing precision in predicting flood-prone regions and aiding in disaster preparedness and response planning (Tepetidis et al. 2025).

The recent progress in AI technologies for wildfire monitoring is enormous. A curious study used deep learning techniques for the fusion of satellite and drone images to ascertain wildfire hotspots, predict spread potential, and provide input on environmental damage (Devi et al. 2024). Another study highlights the integration of drone-captured images with deep learning algorithms for autonomous wildfire detection, achieving over 97% accuracy and over 99% precision using an ensemble approach, significantly enhancing early pile fire detection capabilities compared to traditional methods (Joshi et al. 2024).

Another vital application is the detection of earthquake damage via AI models in disaster management. A study (Ilmak et al. 2024) developed an efficient deep learning-based system, analyzing Maxar's high-spatial-resolution satellite imagery to separate the post-earthquake buildings into classes of collapsed and non-collapsed, thus arming the emergency respondents with an even more efficient way of assessing damage. Another study (Divya et al. 2024) used AI coupled with synthetic aperture radar (SAR) data in the detection of landslide-prone areas, furnishing very useful insights toward regions that highly deserve immediate evacuation efforts and appropriate infrastructure planning.

Storm and hurricane monitoring and impact assessments are also surveyed by AI. Researchers have constructed a hybrid deep learning model to analyze meteorological data from satellite to predict flooding with high accuracy (Abdelkader et al. 2024). Researchers also performed transfer learning on AI trained by historical hurricane data to predict storm surges (Stefanos et al. 2024).

Generally, AI in the field of disaster management, depending on the power it garners from remote sensing, augments emergency preparedness, real-time response, and post-disaster relief. AI models expedite response times, better fitting resource provision, and damage assessment accuracy in coping with diverse nature(scale)-related disasters.

Challenges in AI-Driven remote sensing

Marked by prominent gains in the field of remote sensing, artificial intelligence (AI) allows significant automation features, pattern detection, and data insight into transmission. However, the integration of AI into the processing of satellite images faces challenges. All such challenges impact the growth and use of AI in remote sensing applications. Thus, some of these challenges need to be overcome to utilize AI for optimal benefit in environmental monitoring, disaster management, urban planning, etc. Here, we discuss some of the main challenges implicated in applying AI to remote sensing and satellite-image processing.

Quality and Availability of Data: The availability of high-quality, labeled data remains a core challenge in AI-driven remote sensing (Mingyuan et al. 2024). Many AI-based models, in particular deep learning models, require large, diverse, well-annotated datasets for the training procedure to warrant high accuracy and good generalization (Surbhi et al. 2024). In terms of remote sensing, however, labeled datasets tend to be quite limited, especially in specialized applications or in areas where labeled imagery is simply limited or not available (Maximilian et al. 2024). Moreover, acquiring high-resolution satellite imagery can be costly, with the spatial, spectral, and temporal resolutions from different sensors potentially being so different as to pose challenges to data harmonization (Else et al. 2022). Open access data sharing and synthetic data generation are other ideas, but again, data quality and adherence remain the major challenges.

Computational Complexity: The requirements for deep learning processing in remote sensing are considerable, especially when high-resolution or multispectral imagery containing data points of high volume is concerned. In fact, the training and deployment of these very models require hardware resources of high power, often GPUs or TPUs, which may not be accessible to the majority of organizations,

especially in such resource-constrained settings. Feasibly, these are highly computationally intensive AI models, limiting their application to real-time processing, which remains paramount in disaster responses and the like (Furkan et al. 2024). Model compression and pruning approaches will help alleviate some of that pressure, while the integration of cloud or edge computing must also be considered; nevertheless, semi-efficient real-time processing at a much larger scale remains a significant question.

Interpretability and Transparency: Many AI models, especially deep learning architectures, operate as “black boxes,” making it difficult to understand the decision-making process behind their predictions (Oku et al. 2024). This lack of interpretability presents challenges for remote sensing applications where transparency is crucial, such as disaster management and environmental regulation, where stakeholders need to understand the basis of AI-driven decisions. Explainable AI (XAI) research is developing solutions to these problems through mechanisms that increase the interpretability of AI models. Nevertheless, relatively few solutions are already in the market for actual end-user applications in remote sensing.

Data Fusion and Integration: Remote sensing is often data integration from different sources, such as multispectral, hyperspectral, radar, and thermal sensors, and ancillary data such as weather data or field measurements (Mengmeng et al. 2023). It is the challenge of integrating these diverse types of data into coherent, accurate models that adds to the complexity, owing to disparities in data format, resolution, and acquisition conditions. To derive trustworthy information, AI models would need to learn efficiently from these heterogeneous datasets; however, data fusion is difficult and resource-intensive. Some progress is being made in real-life applications of multimodal learning and sensor fusion techniques, but this process is still evolving, especially in the context of large-scale implementations.

Generalization and Transferability: Generalization is hard for many AI models trained on certain datasets when applying them to new instances or regions, sensors, or environmental conditions (Friedman et al. 2024). For example, a model trained for vegetation pattern assessment in a given area may not do so well in a different ecosystem or climatic setting (Rebecca and James 2024). This predicament thwarts generalization and creates a big hurdle for AI in remote sensing, whereby the applicability of models across various geographic locations or climatic contexts is limited. Transfer learning, self-supervised learning, and domain adaptation techniques appear as a light at the end of the tunnel, allowing the adaptation of the model to the new environment with minimal additional data. Yet, these require maturity for consistent performance through varying remote sensing tasks.

Future directions

There is a bright path ahead for AI in remote sensing, with several avenues for innovative research and technology integration into this field. One future research area would lie in multimodal data fusion, where data from different sensors such as optical, SAR, synthetic aperture radar, and LiDAR are integrated for a more holistic view of the Earth's surface. The fusion of multispectral, hyperspectral, and radar data can substantially increase the accuracy and robustness of the AI models, particularly in environments where a single sensor is inadequate. Thus, AI models performing data fusion across several sensors will provide better precision in monitoring land-use changes, vegetation health, and natural disasters.

Another development along the way is Explainable AI (XAI), and this entails much more. Development of explainable AI models will present stakeholders with a lucid rationale for when and why such models output what they do, for purposes of awesome transparency and winning trust. This is particularly important for critical applications, including disaster response, environmental monitoring, and urban planning, where AI-driven decisions can have significant societal and economic implications. The XAI methods thus developed, such as saliency maps, attention mechanisms, and feature attribution, should all serve to improve the interpretability of the AI models without sacrificing performance.

The amalgamation of AI and edge computing looks to position itself beautifully in the future of remote sensing. Edge computing enables real-time processing of satellite data directly on the sensors or nearby, resulting in faster decision-making due to reduced latency (Jinming et al. 2024). This is particularly useful for any time-sensitive application, such as a natural disaster response and emergency management, where processing the right information in a timely manner can save lives and reduce damage. Advances in lightweight AI models and edge computing hardware will further favor the prompt processing of satellite data on-site and render remote sensing technology better equipped to respond and adapt to changing conditions.

Another aspect that may revolutionize AI-based remote sensing is federated learning. Federated learning allows models to be trained in cooperation from different data centers while keeping raw data siloed, thus preserving data privacy and security (Balamurugan 2024). Federated learning in remote sensing will facilitate cross-border collaborations, allowing institutions from different regions to train models based on datasets without revealing sensitive information. This will not only improve the generalization of models but also culminate in international collaborations on monitoring

such global phenomena as climate change, deforestation, and disaster management.

In addition, advances in transfer learning and self-supervised learning are expected to play a critical role. Transfer learning employs AI models that have been trained in one domain with vast datasets and then fine-tuned for task-specific applications in remote sensing to reduce the amount of labeled training data. As for self-supervised learning, where models are capable of learning from unlabeled data, the technique has proven invaluable in leveraging large amounts of unannotated satellite imagery for downstream improvements in the classification, segmentation, and anomaly detection models' performance (Ali and Hossein 2024).

Another future direction refers to the use of GANs for synthetically generating data. These networks can be used for the creation of realistic yet synthetic satellite images. Such images supplement the existing datasets, hence strengthening the models. They address a drawback of limited training data and contribute to improving diversity in the examples used for AI training, with particular attention to underrepresented regions.

Recently, increasing interest has been realized in combining AI with citizen science and crowd-sourcing data (Shu-Wei et al. 2024; Karin et al. 2024; Sameer and Priyanka 2024). Merging both satellite imagery and observations from the ground by citizen scientists would give a more complete and well-detailed coverage of much more of the environment. Thus, AI models that use remote sensing data and public information can revolutionize environmental monitoring, improve urban planning, and significantly assist during disaster response. In this way, a more comprehensive understanding is gained of the dynamics under which changes occur across the Earth's surface.

Conclusion

The integration of artificial intelligence (AI) in remote sensing and satellite image processing has brought new possibilities across multiple application domains, from environmental monitoring to urban planning, from climate change analysis to disaster management, and so forth. This review highlighted the various AI techniques applied in the field, such as identifying machine learning, deep learning, transfer learning, and data fusion approaches that enable efficient and accurate assessments of the accumulated volume and aggravation of complexities of satellite data. Hence, researchers, practitioners, and regulators will obtain meaningful insights from the data that are otherwise difficult to process manually, thus accelerating decision-making and proactive responses to global challenges.

Though these advancements are highly promising, numerous challenges remain, especially concerning data quality and computational requirements, interpretability of models, and generalizability of an AI model across different geographies and environmental settings. This will require continuous investment in high-quality datasets, more robust computational infrastructure, and the interpretation as well as scalability of models that can take on the ambiguities and complexities of remote sensing data.

In all likelihood, the future of AI in remote sensing will be promising, particularly regarding breakthroughs such as multimodal data fusion, explainable AI, edge computing, and federated learning. This will render AI-aligned remote sensing not only more efficient and effective, but rather, significantly lower the bar to access and actionize it for stakeholders of diverse sectors. The impact that AI models will have on sustainable development, disaster resilience, or environmental stewardship will rise tremendously as the models become increasingly flexible, interpretable, and cooperative.

In summation, AI-driven remote sensing has what it takes to be a major player in addressing global challenges, delivering critical insights, and informing data-driven approaches to complex issues. Interdisciplinary collaboration and technological innovation will remain crucial in future efforts to leverage the full potential of AI in remote sensing and its beneficial impact on society and the environment.

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Declarations

Competing interests The authors declare no competing interests.

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